

KHOLM, Russian FederationWMO: **263780**Lat: **57.150N**Lon: **31.183E**Elev: **71**StdP: **100.47**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-21.4	-27.9	0.3	-24.9	-24.1	0.4	-21.1	6.4	-2.2	5.9	-1.7	0.5	30	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	28.1	19.5	26.5	18.4	25.0	17.8	20.7	25.8	19.7	24.7	18.7	23.3	2.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	23.1	17.9	13.0	21.9	16.9	12.1	20.8	59.8	25.5	56.6	24.9	53.1	23.2	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.3	4.7	-29.5	30.7	2.9	1.4	-31.6	31.7	-33.2	32.5	-34.9	33.3	-37.0	34.4		
(5)			-29.7	22.3	2.9	0.4	-31.7	22.6	-33.4	22.8	-35.0	23.1	-37.1	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-6.8	-3.9	-0.7	5.8	12.0	14.9	17.1	15.9	11.4	5.2	0.4	-3.0	(6)
(7)		DBStd	9.42	7.06	6.36	5.20	4.22	4.28	3.73	3.09	3.21	3.85	4.42	4.48	5.97	(7)
(8)		HDD10.0	2290	522	389	332	139	27	6	0	1	28	155	287	404	(8)
(9)		HDD18.3	4645	781	623	590	377	199	113	60	89	208	406	537	663	(9)
(10)		CDD10.0	738	0	0	0	13	89	153	222	184	71	7	0	0	(10)
(11)		CDD18.3	51	0	0	0	0	4	9	23	14	1	0	0	0	(11)
(12)		CDH23.3	723	0	0	0	3	101	152	274	175	18	0	0	0	(12)
(13)		CDH26.7	120	0	0	0	0	19	21	51	28	1	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.3	2.4	2.7	2.3	2.2	2.1	1.8	1.9	2.0	2.4	2.5	2.6	(14)
(15)	Precipitation	PrecAvg	747	56	42	38	45	61	79	103	81	69	63	62	53	(15)
(16)		PrecMax	978	83	71	76	103	100	151	200	150	148	132	114	94	(16)
(17)		PrecMin	581	21	8	10	12	25	10	13	15	18	19	0	29	(17)
(18)		PrecStd	103	20	15	22	26	22	41	61	42	36	29	32	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	5.7	14.2	23.6	29.0	29.1	30.0	29.1	25.5	17.2	11.2	8.7	(19)
(20)			MCWB	4.6	3.2	8.1	12.2	18.4	18.5	20.4	19.8	18.4	13.0	10.5	7.4	(20)
(21)		2%	DB	3.8	4.0	10.2	20.1	26.1	26.7	28.0	27.3	23.0	14.3	9.2	6.2	(21)
(22)			MCWB	3.0	2.2	5.1	11.5	16.4	18.3	19.8	19.0	17.6	11.3	8.2	5.3	(22)
(23)		5%	DB	2.6	3.2	7.7	16.7	23.6	25.1	26.4	25.4	20.6	12.8	7.7	4.9	(23)
(24)			MCWB	1.6	1.7	3.9	9.7	15.1	17.3	19.0	18.6	15.9	10.8	6.7	4.1	(24)
(25)		10%	DB	1.5	2.4	5.7	14.3	21.1	23.3	24.6	23.3	18.5	11.8	6.2	3.2	(25)
(26)			MCWB	0.7	1.2	3.2	8.3	13.6	16.3	18.4	17.6	14.4	10.2	5.3	2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	3.7	8.6	14.5	19.7	21.6	21.7	21.5	19.2	13.5	10.8	7.6	(27)
(28)			MCDWB	5.3	5.1	13.4	20.4	26.9	26.0	26.8	26.6	24.4	15.8	10.9	8.7	(28)
(29)		2%	WB	3.1	2.6	6.0	12.1	17.8	20.0	20.8	20.2	17.6	12.0	8.5	5.5	(29)
(30)			MCDWB	3.7	3.6	8.2	17.7	23.7	24.9	26.2	25.1	22.0	13.4	9.1	6.0	(30)
(31)		5%	WB	1.8	2.0	4.5	10.5	16.4	18.4	19.9	19.3	16.4	11.2	6.7	4.1	(31)
(32)			MCDWB	2.5	2.9	6.8	16.3	21.3	22.5	24.9	23.9	20.0	12.6	7.5	4.9	(32)
(33)		10%	WB	0.8	1.4	3.4	8.8	15.3	17.2	18.7	18.1	15.1	10.3	5.4	2.5	(33)
(34)			MCDWB	1.5	2.2	5.6	13.4	19.4	21.7	23.4	22.3	18.0	11.9	6.2	3.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	6.4	8.3	11.2	13.3	12.4	11.0	12.3	10.4	6.8	4.3	4.1	(35)
(36)			MCDBR	3.8	5.5	11.3	16.9	17.7	15.9	15.2	16.0	15.0	9.0	5.3	3.7	(36)
(37)			MCWBR	3.5	4.2	7.2	9.5	8.6	8.0	7.5	8.5	9.3	6.8	4.8	3.4	(37)
(38)		5% WB	MCDWB	3.7	3.9	9.3	14.8	14.3	13.1	13.5	13.8	13.5	8.2	5.1	3.4	(38)
(39)			MCWBR	3.5	3.3	6.4	8.6	7.2	7.1	7.1	7.9	8.5	6.5	4.8	3.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.321	0.343	0.389	0.380	0.374	0.398	0.403	0.366	0.345	0.325	0.283	(40)	
(41)		taud	2.139	2.122	2.175	2.254	2.364	2.403	2.350	2.330	2.408	2.382	2.240	2.149	(41)	
(42)		Ebn at Noon	565	702	799	821	857	867	833	800	779	687	518	459	(42)	
(43)		Edh at Noon	55	86	107	116	111	108	112	107	86	68	51	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.44	1.23	2.65	3.89	5.18	5.43	5.25	4.19	2.65	1.18	0.40	0.25	(44)	
(45)		RadStd	0.10	0.19	0.34	0.47	0.50	0.49	0.54	0.39	0.29	0.23	0.07	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	-158	-181	N/A	N/A

Nomenclature: See separate page